TEXORA · LECTURE SHEET

Ergonomics

Ergonomics fits the workplace to the worker instead of forcing the worker to fit the workplace. In a mill it decides whether a weaver can patrol her looms all shift without injury, whether an inspector can actually see the fault, and whether the air itself is fit to breathe. This sheet covers definition, importance, workplace and layout application, ventilation, filtration and mill illumination — with a full lighting design calculation.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 10 · LEVEL 4 TERM 1

1. Definition and scope

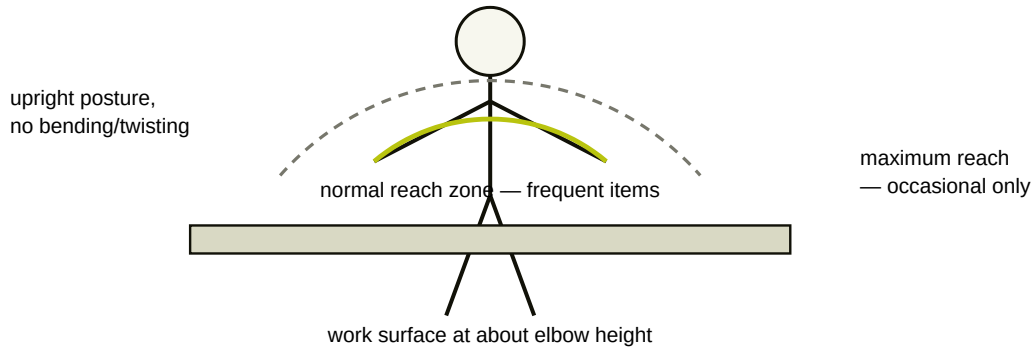
Ergonomics (human factors engineering) is the scientific study of the interaction between people and their work system — machine, task, environment and organisation — applied to design workplaces that are safe, comfortable and productive. It draws on **anthropometry** (body dimensions), **biomechanics** (forces and postures), physiology (fatigue, heat stress) and psychology (displays, controls, error). In the mill its questions are concrete: bench and machine heights, reach distances for cans and beams, lifting loads (Topic 5), patrol routes, noise, heat, dust — and the environmental trio of this syllabus: **ventilation, filtration and illumination**.

2. Importance

- **Health:** badly designed work produces musculoskeletal disorders (back, neck, shoulder and wrist injuries), eye strain, hearing and respiratory harm — the slow accidents of Topic 11.
- **Productivity and quality:** a worker fighting glare, heat or an awkward reach makes errors: missed faults at inspection, wrong settings, damaged material. Comfortable workers sustain attention all shift.
- **Safety:** fatigue and poor visibility are root causes behind a large share of accidents; ergonomic design is preventive safety engineering.
- **Economics:** absenteeism, turnover and compensation fall; the cost of correct bench heights and lighting is trivial beside one lost-time injury or a claims lot of faulty cloth.

3. Application to workplace and layout

- **Work height:** working surfaces near elbow height for light work; lower for heavy force work, slightly higher for fine visual work. Adjustable stands serve mixed-height workforces.
- **Reach zones:** frequent items within the normal reach envelope (forearm sweep), occasional items within maximum reach, nothing frequent above shoulder or behind the body (Figure 1).
- **Posture:** alternate sitting and standing; provide sit-stand stools on patrol jobs; avoid sustained bending/twisting — bring the work to the worker (tilting inspection frames, roll lifters).
- **Manual handling:** apply Topic 5 mechanisation; set load and frequency limits for residual manual tasks; team lifts and trolleys for beams and rolls.
- **Layout:** machines arranged for straight material flow with clear gangways (marked, unobstructed), services overhead or in trenches — not across walkways; noisy and dusty processes separated or enclosed; rest, water and first-aid points within reach of every section.
- **Controls and displays:** controls within reach, moving in the direction the operator expects; critical gauges readable from the working position; emergency stops prominent (link to Topic 6 automation and Topic 11).



Workstation ergonomics: posture upright, surface at elbow height, frequent items inside the normal reach arc.

Figure 1 — Workstation ergonomics sketch. The solid arc is the normal reach zone for frequent items; the dashed arc is maximum reach, for occasional items only.

4. Ventilation

Ventilation supplies fresh air and removes contaminated, hot or humid air to keep the workplace healthy and the process stable (humidity control itself is Topic 1). Mill needs: dilute fibre dust and chemical vapours below occupational exposure limits, remove machine heat, and supply make-up air for exhaust systems so they actually work.

- **General (dilution) ventilation:** whole-room air changes by fans/ducted plant — the air-washer system of Topic 1 doubles as the mill's ventilation.

- **Local exhaust ventilation (LEV):** hoods capture dust/vapour at the source — blowroom and carding extraction, dye-kitchen hoods, stenter exhausts — far more effective per m^3 of air than dilution.
- **Design checks:** air must flow from clean toward dirty zones; intakes sited away from exhausts and boiler stacks; supply and exhaust balanced so doors and dampers behave.

5. Filtration

Exhausted and recirculated air must be cleaned — to protect workers, product and neighbours. **Filtration** stages in textile service:

- **Pre-filters (coarse):** panels and rotary air filters removing lint and coarse dust from air-washer intake and recirculation — protect later stages and coils.
- **Fine filters:** bag/pocket filters for supply air to sensitive departments and testing laboratories.
- **Cyclones and dust plants:** blowroom/card waste extraction separates fibre and dust by centrifugal action into compactors — collection, not filtration, but part of the same air system.
- **Activated carbon / scrubbers:** for chemical vapours and odour where wet processing requires them.

Filters are maintenance items (Topic 9): a choked filter silently cuts airflow, raises fan energy and lets dust bypass — pressure-drop gauges and change schedules are the control.

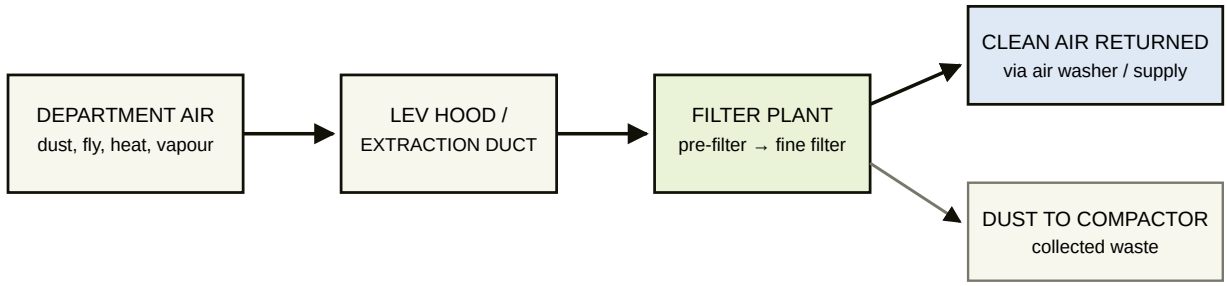


Figure 2 — Ventilation and filtration airflow. Local exhaust captures contaminated air at the source; filtration returns clean air and removes dust as collected waste.

6. Mill illumination

Illumination design sets the light level (illuminance, in **lux** = lumens per m^2) that each task needs: rough circulation areas need least; production floors more; inspection and shade matching need most — and also demand good **colour rendering** and freedom from glare and flicker. Design balances quantity (lux on the task), quality (uniformity, glare control, colour rendering index), daylight use, and energy

(efficient luminaires, occupancy/daylight controls). Typical design targets used in mill practice (illustrative) are of the order of 200–300 lux for weaving/spinning floors and 500 lux or more at inspection — the exact values must be taken from the applicable standard for the project.

WORKED EXAMPLE — LUMEN METHOD OF LIGHTING DESIGN

Hall 30 m × 20 m → area A = 600 m². Target illuminance E = 300 lux. Each luminaire emits Φ = 4,000 lm. Utilisation factor UF = 0.5 (fraction of emitted light reaching the working plane). Maintenance factor MF = 0.8 (allowance for dirt and lamp ageing).

Formula: N = (E × A) / (Φ × UF × MF).

N = (300 × 600) / (4000 × 0.5 × 0.8) = 180,000 / 1,600 = **112.5 → 113 luminaires**.

Layout check: 113 luminaires over 600 m² is one per 5.3 m² — e.g. a grid of about 9 × 13 = 117 fittings at roughly 2.3 m × 2.3 m spacing would satisfy the count with a regular pattern; the designer adopts a buildable grid at or above the calculated number, never below.

The maintenance factor is the ergonomic link to Topic 9: a design that ignores dirt depreciation delivers its design lux only on the day of installation. Cleaning schedules for luminaires and light-coloured, clean surfaces (high room reflectances raise UF) are part of the lighting system.

Glossary

Ergonomics — designing work to fit human capabilities and limits. **Anthropometry** — measurement of body dimensions used in design. **Reach zone** — the area a worker can reach without stretching or bending. **LEV** — local exhaust ventilation capturing contaminants at source. **Lux** — illuminance unit, lumens per square metre. **Utilisation factor** — fraction of lamp lumens reaching the working plane. **Maintenance factor** — allowance for dirt and ageing in lighting design. **Glare** — excessive brightness contrast that reduces seeing and comfort.

Key points

• Ergonomics = fit the job to the worker: posture, reach, load, environment.

• Ventilation dilutes and removes contaminants; LEV captures them at source; filtration cleans recirculated and exhaust air.

• Lumen method: N = EA/(Φ·UF·MF) = 180,000/1,600 = 112.5 → 113 luminaires for the 600 m², 300 lux hall.

• Lighting quality (uniformity, glare, colour rendering) matters as much as lux — above all at inspection.

Practice

1. A hall of 40 m × 15 m needs 250 lux. Luminaires emit 5,000 lm, UF = 0.55, MF = 0.8. Calculate the number required.
2. An inspector reports eye strain and missed faults though a lux meter shows the design level at bench centre. Give three non-lux causes to check.
3. Explain why LEV is preferred over general dilution ventilation for carding dust, in energy and health terms.
4. List four layout changes that improve ergonomics in a weaving shed without buying new looms.

Answer key

1. A = 600 m². N = (250 × 600)/(5000 × 0.55 × 0.8) = 150,000/2,200 = 68.2 → **69 luminaires**.
2. Any three: glare from unshielded fittings or windows; poor uniformity (bright/dark patches); flicker; poor colour rendering hiding shade faults; light from the wrong direction (specular reflection off the cloth); dirty luminaires away from the measured point (low real MF).
3. LEV captures dust at the card before it mixes into room air, so far less air volume must be moved and cleaned for the same exposure reduction — less fan energy — and the worker's breathing zone is protected directly rather than on average.
4. Any four: raise/lower working surfaces to elbow height; relocate frequent items into normal reach; add sit-stand stools and anti-fatigue mats; clear and mark gangways; move services off walkways; add roll/beam lifting aids; reposition gauges and stops within reach and sight.

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